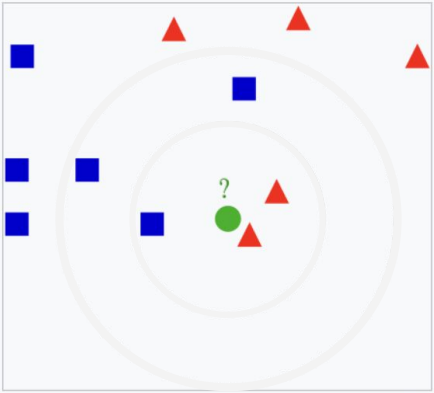
Our first ML Model

- k-Nearest neighbors (k-NN)
  - Supervised learning
  - Non parametric ***(Distance based method)***
  - Both for Classification and Regression solutions



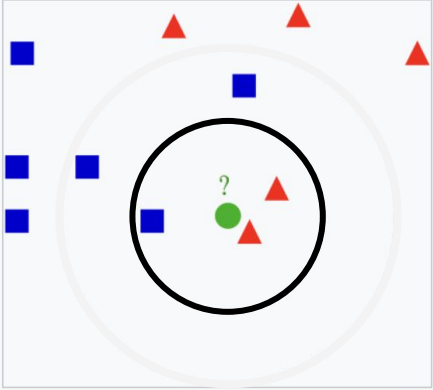
Example of *k*-NN classification. The test sample (green dot) should be classified either to blue squares or to red triangles. If *k = 3* (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If *k = 5* (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).

src: https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

Our first ML Model

- k-Nearest neighbors (k-NN)
  - Supervised learning
  - Non parametric ***(Distance based method)***
  - Both for Classification and Regression solutions

- For
  - *k = 3, and*
  - *Using L2/Euclidean distance as proximity metric*

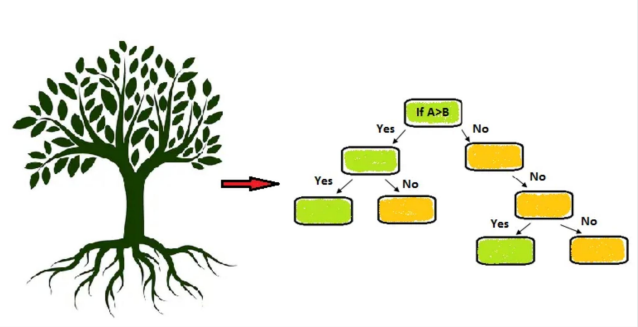


Example of *k*-NN classification. The test sample (green dot) should be classified either to blue squares or to red triangles. If *k = 3* (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If *k = 5* (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).

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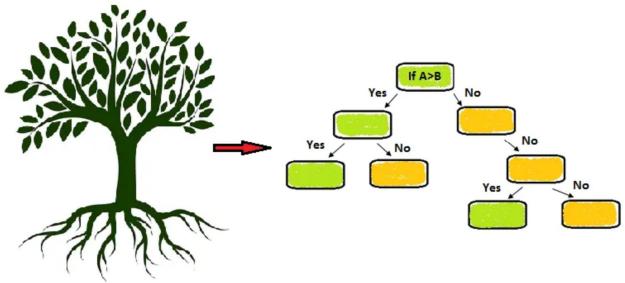
CIS 635 Knowledge Discovery & Data Mining

ML Models: Decision Tree



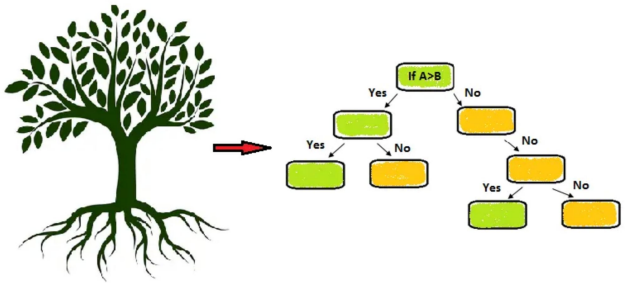
Decision Tree

- Another non-parametric model
  - Recall k-NN, its an in memory model; right?



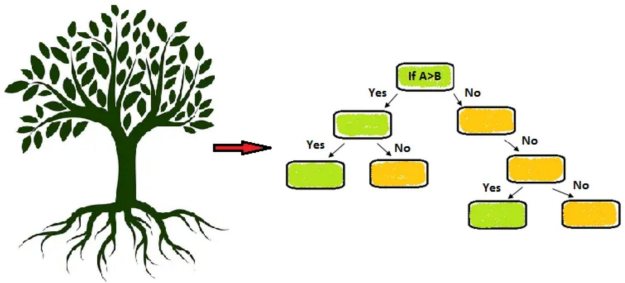
Decision Tree

- Another non-parametric model
  - Recall k-NN, its an in memory model; right?
- Decision Tree is our second example



Decision Tree

- *Concepts and Principles*
- *Let's learn through an example*

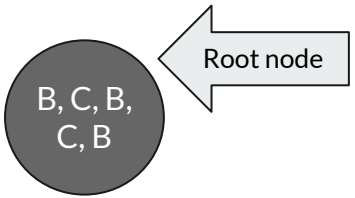


Decision Tree

- Data records for two animal classes:
  - **Bunny** and **Cat**

nb of legs	weight (lb)	animal
4	2.1	Bunny
4	7	Cat
4	1.7	Bunny
4	9	Cat
4	2.75	Bunny

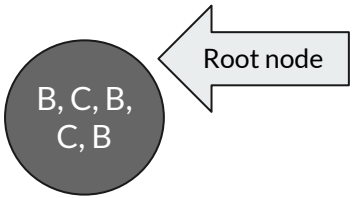
Decision Tree



- What feature should we use to split records?

nb of legs	weight (lb)	animal
4	2.1	Bunny
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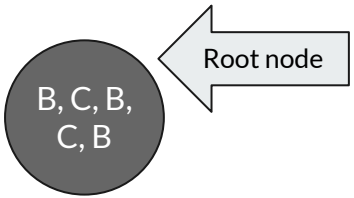
Decision Tree



- What feature should we use to split records?
- nb of legs is useless as there is no variation.

nb of legs	weight (lb)	animal
4	2.1	Bunny
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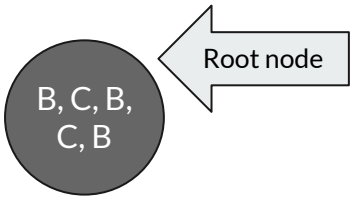
Decision Tree



- *What feature should we use to split records?*
- *nb of legs is useless as there is no variation.*
- *We can use the '**weight(lb)**' feature.*

nb of legs	weight (lb)	animal
4	2.1	Bunny
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4	9	Cat
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Decision Tree

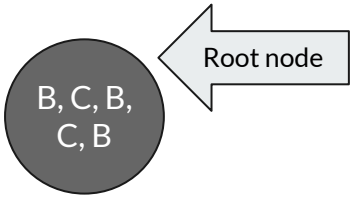


- *Let's plot the data points*

nb of legs	weight (lb)	animal
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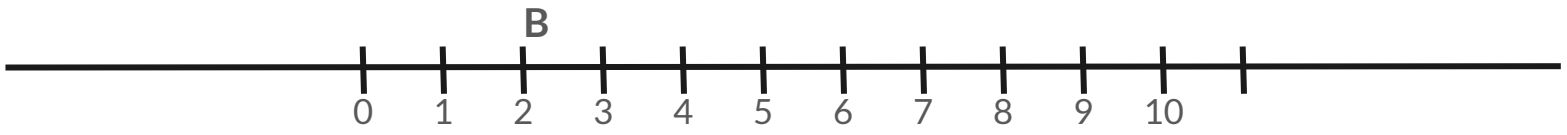


Decision Tree

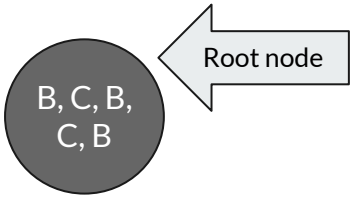


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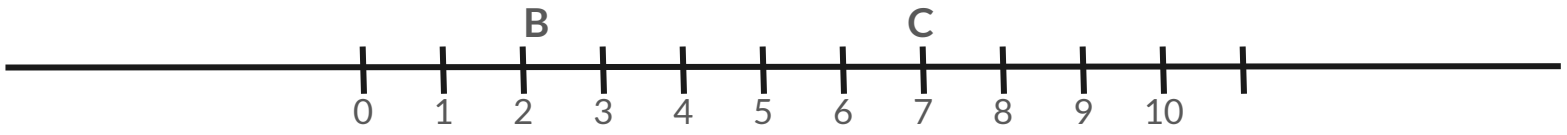


Decision Tree

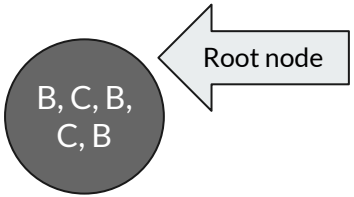


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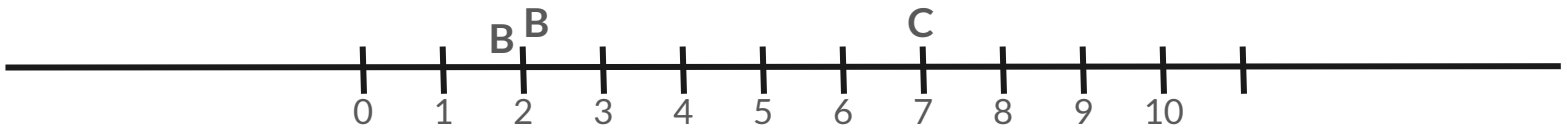


Decision Tree

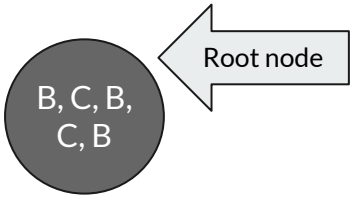


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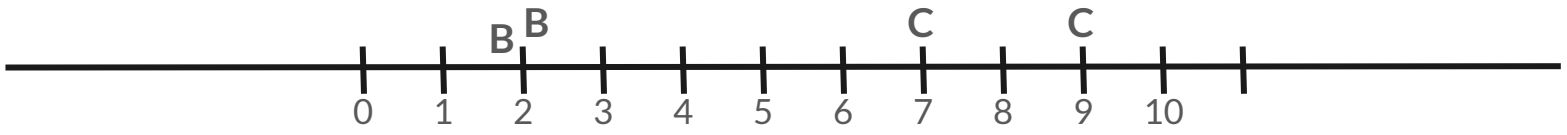


Decision Tree

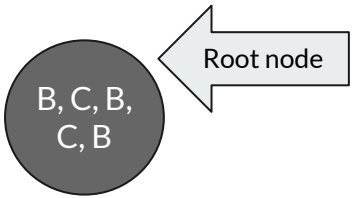


- *Let's plot the data points*

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Decision Tree

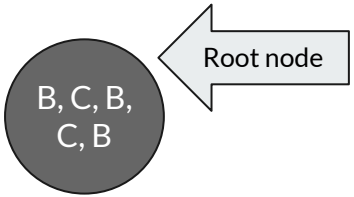


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Decision Tree

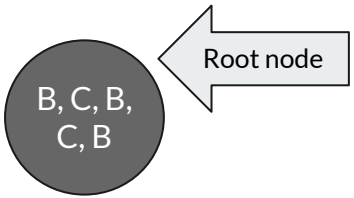


nb of legs	weight (lb)	animal
4	2.1	Bunny
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4	1.7	Bunny
4	9	Cat
4	2.75	Bunny

- *Can we identify groups?*

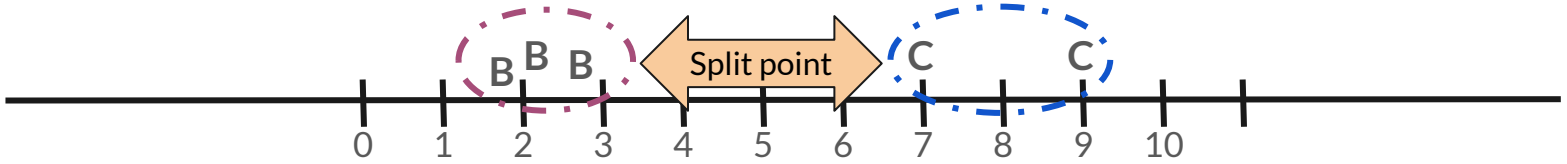


Decision Tree

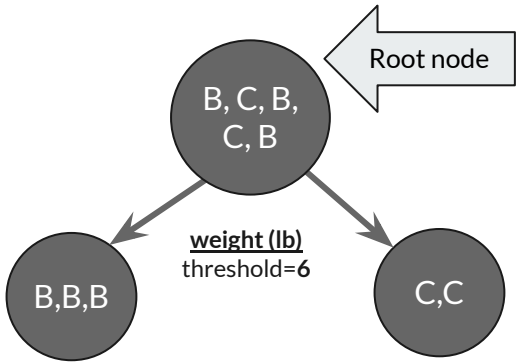


- *Let's find a split point.*

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4	2.75	Bunny

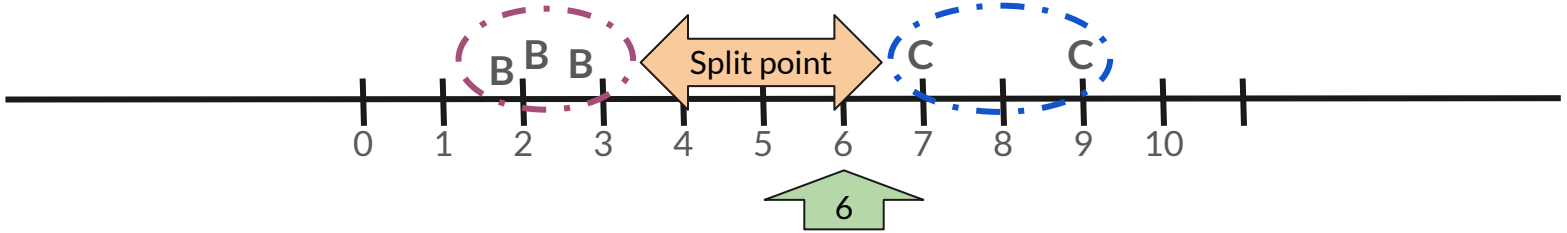


Decision Tree

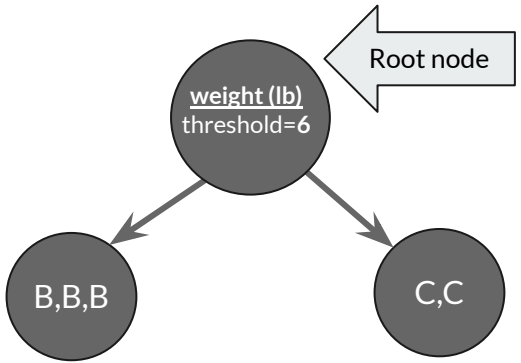


nb of legs	weight (lb)	animal
4	2.1	Bunny
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4	1.7	Bunny
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- *Create branches*

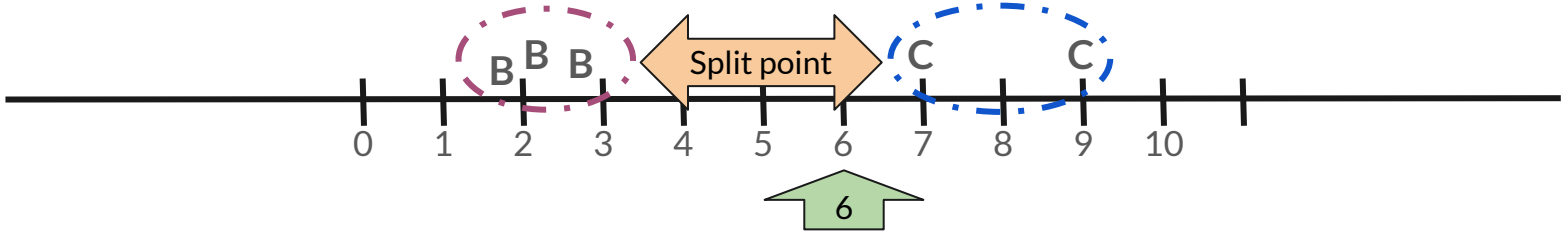


Decision Tree

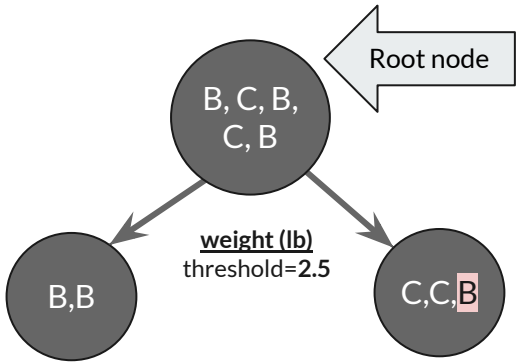


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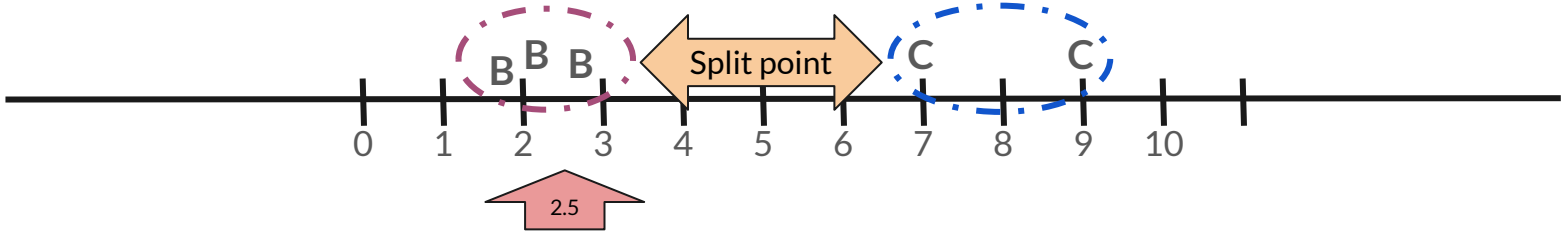


Decision Tree

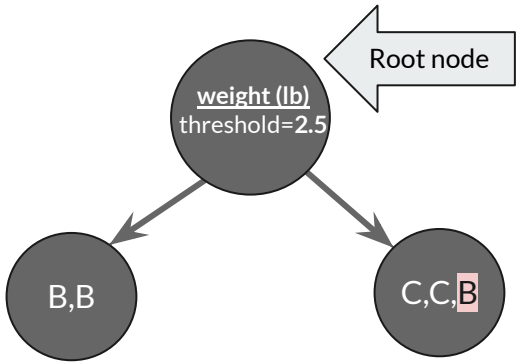


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- *What if we used a threshold=2.5?*

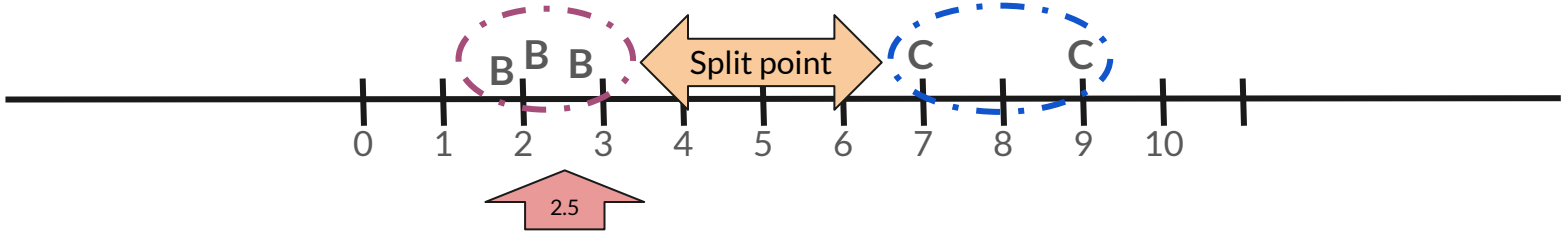


Decision Tree



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- *What if we used a threshold=2.5?*



Decision Tree

- What metrics are used for split point determination?

Decision Tree

- What metrics are used for split point determination?

Entropy

Gini Impurity

Information Gain

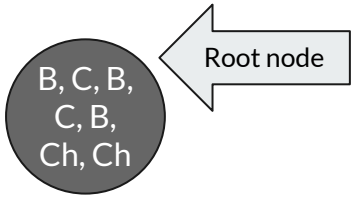
The idea is quite simple, choose the one that make classes more separable.

Decision Tree

- How about this configuration?
- We have data points for an additional animal class “**Chicken**”

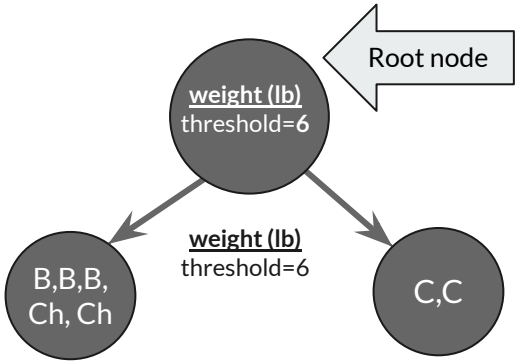
nb of legs	weight (lb)	animal
4	2.1	Bunny
4	7	Cat
4	1.7	Bunny
4	9	Cat
4	2.75	Bunny
2	2.5	Chicken
2	3	Chicken

Decision Tree



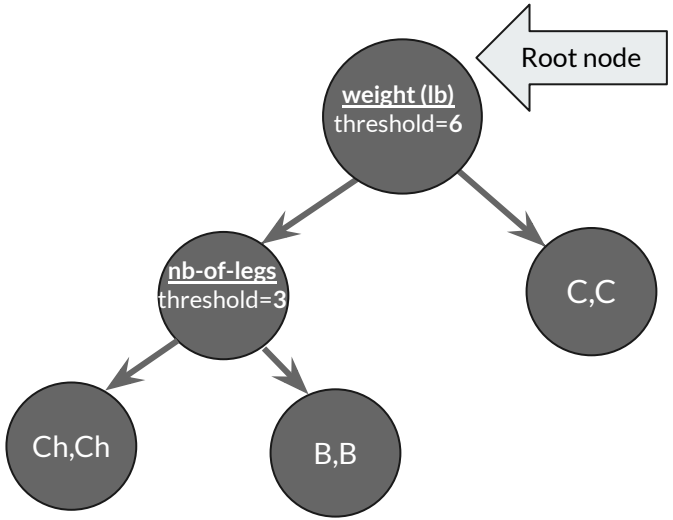
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Decision Tree



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Decision Tree

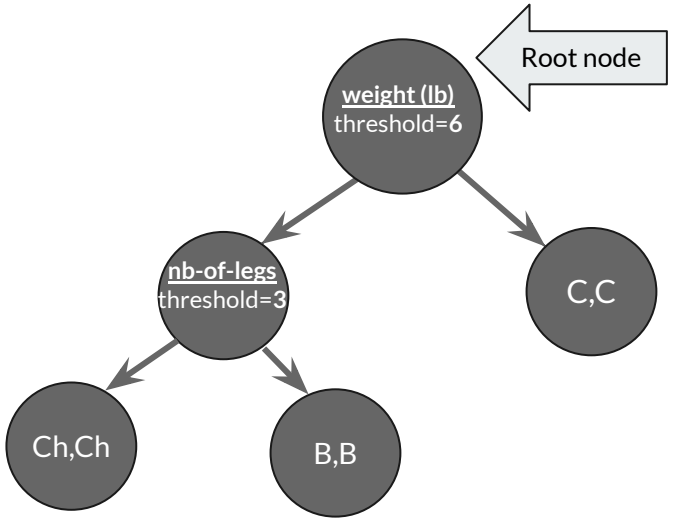


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2	2.5	Chicken
2	3	Chicken

Decision Tree

- Prediction Time

Decision Tree

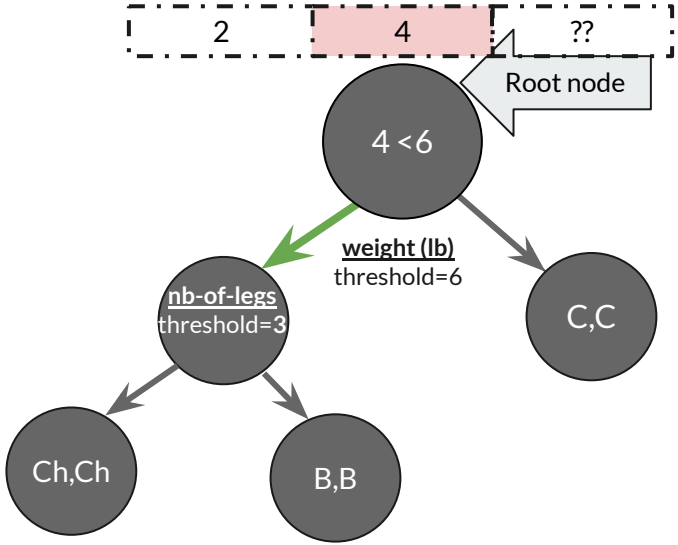


nb of legs	weight (lb)	animal
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4	1.7	Bunny
4	9	Cat
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Test case

2	4	??

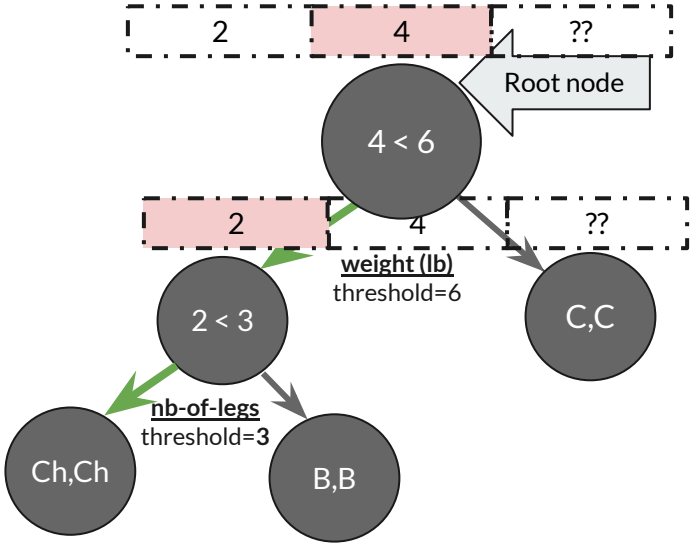
Decision Tree



Test case

2	4	??

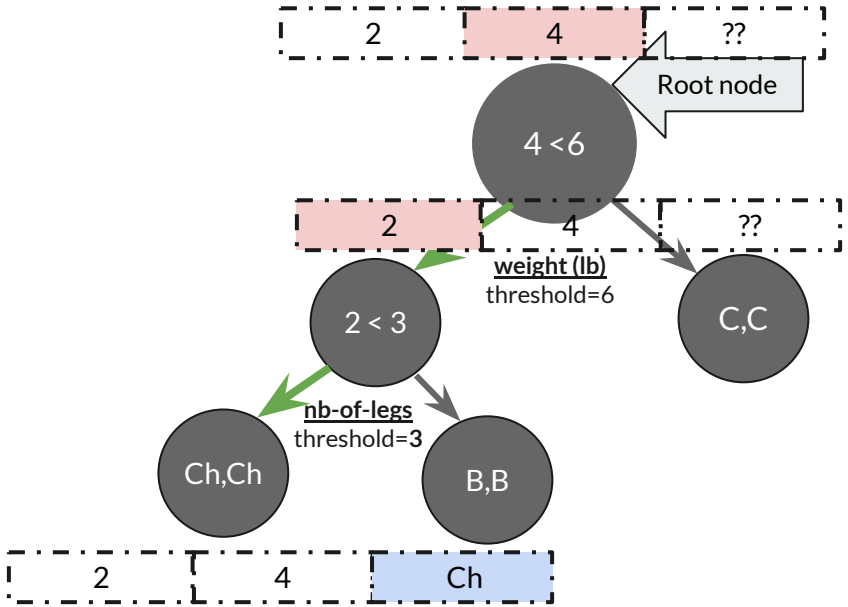
Decision Tree



Test case

2	4	??

Decision Tree



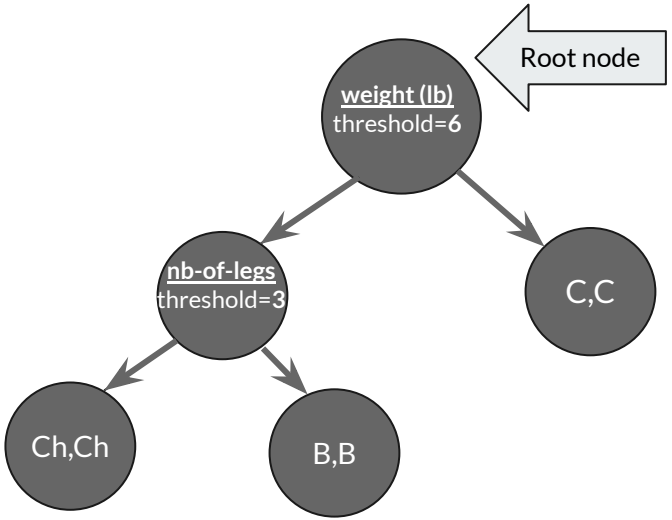
Test case

2	4	??

Decision Tree

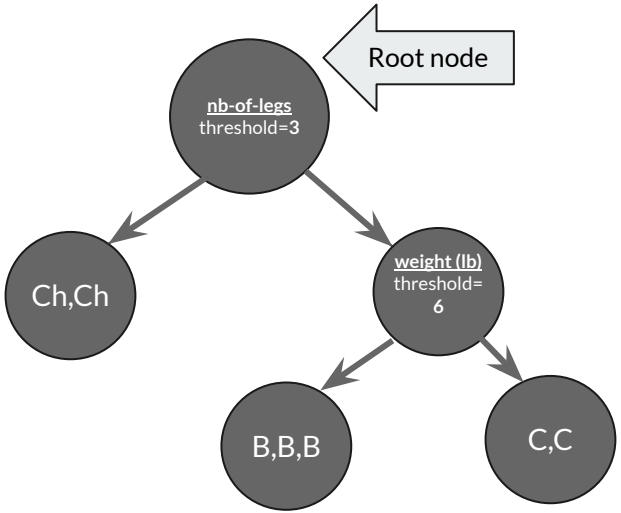
You may have multiple trees

Decision Tree



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Decision Tree



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Decision Tree

Entropy and Information Gain

Decision Tree

Entropy

- What metrics are used for split point determination?

Entropy (discrete variable) :

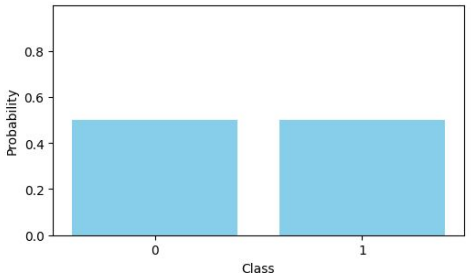
$$H(X) = -\sum_{i=1}^{n} p(x_i) \log_b p(x_i)$$

Decision Tree

- What metrics are used for split point determination?

Entropy

H=1



Entropy (discrete variable) :

$$H(X) = -\sum_{i=1}^{n} p(x_i) \log_b p(x_i)$$

Decision Tree

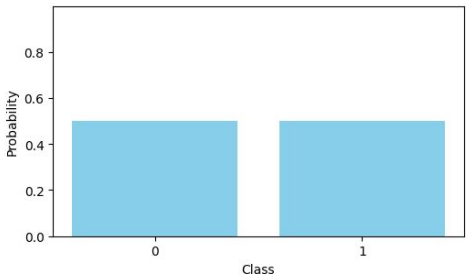
- What metrics are used for split point determination?

Entropy

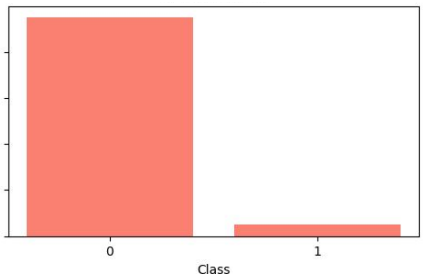
Entropy (discrete variable) :

$$H(X) = -\sum_{i=1}^{n} p(x_i) \log_b p(x_i)$$

H=1



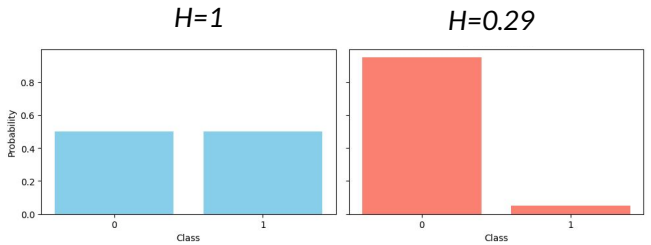
H=0.29



Decision Tree

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Entropy



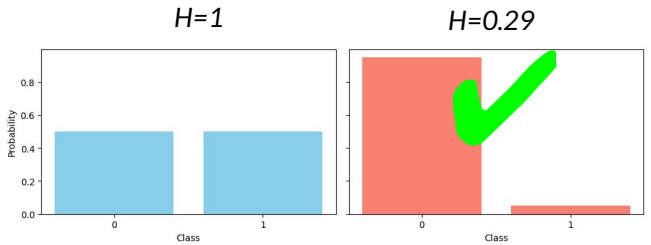
$$IG(T, a) = \mathrm{H}(T) - \mathrm{H}(T|a),$$

where $\mathrm{H}(T|a)$ is the conditional entropy of T given the value of attribute a.

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- Random sampling of data points
- Random sampling of features
- Randomizing feature combinations

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This will also allow us to build advanced models: Random Forest

QA